

Conor O'Neil

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Summary

Principal Mechanical Engineer with over 12 years of experience developing robotic and advanced automation equipment from concept through production. Technical lead and inventor on four published patent applications with expertise in research and development, hardware design, integration, and deployment. Recently relocated from Boston to the Tampa Bay area to support my wife's surgical career.

Professional Experience

PRINCIPAL MECHANICAL ENGINEER | BERKSHIRE GREY/SOFTBANK | MAY 2021 – JULY 2026

- Led hardware engineering development of the [Scoop Robotic Package Unloader](#) from product inception to commercial deployment, resulting in multi-million-dollar contracts from Fortune 50 customers
- Collaborated directly with marketing, key customers, and cross-functional engineering teams to analyze package handling operations in the field, define customer requirements, establish technical scope, and shape the product architecture
- Invented core technologies in suction gripping, vacuum generation, nose roller scooping, and articulating conveyor systems, resulting in four published U.S. and international patent applications with additional applications pending
- Owned the design of structural weldments, sheet metal assemblies, actuation mechanisms, power transmission, sensor integration, and custom conveyors as both a technical lead and hands-on individual contributor for a product with a \$500,000+ BOM and thousands of unique components
- Directed mechanical, electrical, controls, perception, and robotics teams for end-effector design, grasping strategies, force-torque sensing, and conveyor logic, leading design reviews and mentoring engineers

SENIOR MECHANICAL ENGINEER | COGNEX | JULY 2016 – MAY 2021

- Developed autonomous calibration and test equipment for Cognex DataMan barcode readers, In-Sight vision systems, and [3D vision sensors](#) throughout the product lifecycle, from early alpha builds through full-scale production
- Designed and prototyped optomechanical automation equipment, including a motorized dual-diaphragm 5-DoF flexure enabling sub-micron focusing repeatability of C-mount lenses
- Created tolerance analyses and applied GD&T methodologies to achieve $\pm 50 \mu\text{m}$ optical lens-to-image-sensor alignment across multiple calibration machines
- Piloted an NPI initiative to transition calibration and test equipment manufacturing from in-house production to a contract manufacturer, projected to shift approximately \$1 million in annual equipment costs within two years

MANUFACTURING ENGINEER | COGNEX | JUNE 2014 – JULY 2016

- Owned manufacturing, calibration, and test engineering for the flagship DM150/260 DataMan barcode reader platform from development through commercial launch, delivering Cognex's highest-selling product at the time
- Designed and built all production fixtures, including an interchangeable press-fixture platform adopted across multiple assembly lines
- Investigated electrical, PCBA, and field failures through root cause analysis, implementing corrective actions and improving test coverage to increase product quality and manufacturing yield
- Transferred manufacturing of the DM150/260 platform to a contract manufacturer in Indonesia, qualifying test and calibration equipment, training manufacturing teams, and supporting ramp-up to approximately 100,000 readers annually

MECHANICAL DESIGN ENGINEER INTERN | BOSE | SUMMER 2013

- Designed and developed manufacturing equipment for Bose audio products, including a rotating pneumatic motor stand for the [Bose Ride System](#) and a tinsel spreader for transducer assembly

Education

B.S. MECHANICAL ENGINEERING | UNIVERSITY OF MASSACHUSETTS AMHERST | 2014

- GPA – 3.64 – **Cum Laude**
- Dean's List 2010, 2012, 2013, and 2014

Patents

- Vacuum Control Systems and Methods for Use in Object Processing ([US20240326264](#))
- Systems and Methods for Processing Objects Using an Articulated Unloader ([US20240327144](#))
- Systems and Methods for Processing Objects Using Kicker Rollers ([US20240327143](#))
- Systems and Methods for Dynamic Processing of Objects Using Intermediate Storage Caches ([W02024254391](#))

Skills and Projects

CAD & Product Visualization – SolidWorks, PDM/EPDM, SolidWorks Visualize, Keyshot

Engineering Methods – FEA, GD&T, DFM, tolerance stackups, statistical methods, motion analysis

Fabrication – CNC machining, manual mills and lathes, sheet metal fabrication, 3D printing, laser cutting

Metrology – CMMs, portable CMM arms, precision metrology equipment

SENIOR DESIGN PROJECT | UNIVERSITY OF UMASS AMHERST | FALL 2013

- Awarded 2nd place out of 17 teams at the Mechanical Engineering Design Showcase for designing and fabricating a mobile aluminum chassis for a hardwood flooring nail gun, increasing flooring installation speed by up to 33% while reducing operator fatigue

PERSONAL ENGINEERING PROJECTS

- Conceptualized and created a transportable enclosed CNC machine, including custom mechanical gantry design, electrical integration, part fabrication, and precision machine calibration